Sentence-BERT–Driven Semantic Scoring Framework for Subjective Exam Answers

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***Abstract*—Subjective evaluation of descriptive answers in educational environments often requires significant time and may suffer from inconsistencies or unintentional bias, particularly when instructors handle large volumes of answer scripts. To address this challenge, this work proposes a semantic-based scoring approach that employs Sentence-BERT embeddings to assess the conceptual similarity between student responses and predefined reference answers, rather than relying solely on keyword matching. The proposed system computes a normalized similarity value for each response, maps this value to marks using adjustable scoring criteria, and generates concise feedback indicators to support transparent and consistent assessment. Experimental implementation within a web-based evaluation platform demonstrates that the proposed method can substantially reduce manual grading effort while enhancing fairness and reliability in subjective answer assessment..**

Keywords: Semantic similarity; Sentence BERT; Subjective answer evaluation; Automated grading; Natural language processing.

I. INTRODUCTION

Subjective assessment remains a critical component of educational evaluation, as it enables the measurement of advanced cognitive abilities such as conceptual understanding, analytical reasoning, and depth of knowledge. Descriptive examinations are particularly valuable for assessing learning outcomes that cannot be effectively measured through objective or multiple-choice formats. However, the evaluation of such responses involves analyzing multiple qualitative factors, including correctness, clarity, logical flow, and completeness. Consequently, manual grading becomes time-intensive and cognitively demanding, especially when instructors are required to evaluate a large number of answer scripts. Under such conditions, challenges such as evaluator fatigue, scoring variability, and unintended bias may emerge. The rapid growth of online examinations and digital learning environments has further amplified these concerns, emphasizing the demand for automated and scalable evaluation mechanisms.

Initial approaches to automating subjective answer assessment primarily employed keyword-based matching techniques and rule-driven scoring schemes. Although these methods can identify the presence of specific terms, they often fail to capture the underlying meaning of a student's response. Correct answers expressed through alternative phrasing, synonymous terms, or diverse writing styles may be incorrectly penalized. As a result, keyword-centric systems typically suffer from limited accuracy and poor generalization across different disciplines and question structures. Recent progress in natural language processing (NLP), particularly through deep learning and transformer-based architectures, has enabled more effective modeling of semantic information within textual data.

Sentence-BERT (SBERT), an extension of the BERT framework, is designed to generate fixed-length vector representations that encode the semantic content of entire sentences. It has shown strong performance in applications such as semantic similarity computation, paraphrase identification, and information retrieval. By focusing on meaning rather than surface-level lexical patterns, SBERT provides a reliable basis for interpreting descriptive answers. Measuring the semantic similarity between student responses and instructor-defined reference answers allows the system to approximate human-like judgment.

In this work, a semantic scoring framework based on Sentence-BERT is proposed for automating subjective answer evaluation. The system computes similarity measures using SBERT embeddings, converts these values into scores through predefined grading thresholds, and produces brief feedback indicators to support transparency

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in assessment. A prototype implementation within a web-based examination system validates the feasibility of the proposed approach. Experimental observations suggest that the framework significantly reduces manual evaluation effort while improving consistency and fairness in scoring.

II. Related Work

Recent advances in transformer-based language models have significantly influenced automated text understanding tasks, including semantic similarity measurement and subjective answer evaluation. Traditional grading approaches based on keyword matching and statistical overlap often fail to recognize conceptually correct answers expressed using varied vocabulary or sentence structures. To overcome these limitations, contextual embedding models such as BERT and Sentence-BERT (SBERT) have been increasingly adopted to capture deeper semantic relationships within text.

Several studies have specifically explored SBERT for automated answer grading. Das and Chakraborty [1] proposed an SBERT-driven framework for short answer evaluation, where semantic similarity between student responses and reference answers was used as the primary grading signal. Their results demonstrated improved accuracy and robustness compared to lexical matching methods. Similarly, Metan et al. [5] presented an automated subjective evaluation system combining machine learning and NLP techniques, highlighting that semantic similarity-based scoring leads to greater consistency with human assessment. In contrast, Kapparad [8] employed keyword similarity and regression-based techniques, which performed adequately for structured responses but showed limitations when answers conveyed equivalent meaning using different phrasing. Devi et al. [10] further validated that BERT-based contextual embeddings are more effective than conventional NLP methods in capturing intent and semantic correctness in subjective answers.

Beyond educational assessment, SBERT has been successfully applied in various semantic similarity–driven tasks. Tohma et al. [2] utilized SBERT in a hybrid framework for cognate identification across Turkish and Uzbek, demonstrating its effectiveness in modeling cross-lingual semantic relationships. Tar and Htun [3] enhanced SBERT embeddings with ontology-based knowledge for document clustering, showing that incorporating structured semantic information improves clustering accuracy. Lin and Shen [7] proposed a multi-dimensional SBERT similarity fusion approach for patent plagiarism detection, reinforcing the suitability of SBERT for similarity-centric applications that require fine-grained semantic discrimination. Additionally, Uddin et al. [6] introduced Bangla SBERT using multilingual knowledge distillation, enabling reliable sentence embeddings for low-resource languages and expanding SBERT's applicability beyond English.

BERT-based architectures have also been widely adapted to domain-specific NLP problems. Manzini et al. [4] developed a BERT-based model for analyzing electronic health records of diabetic patients, illustrating the adaptability of transformer models to complex, real-world textual data. Liu et al. [9] proposed an enhanced BERT pre-training strategy for Chinese text classification, achieving superior semantic feature extraction compared to standard BERT. Other notable applications include sentiment analysis using hybrid BERT–CNN models [15], extractive summarization of scholarly articles [12], edge-based fake news detection [13], conversational question-answering chatbots [14], and language modeling for Bahasa Indonesia [11]. Collectively, these studies demonstrate the versatility and effectiveness of BERT and SBERT models in capturing semantic meaning across diverse domains.

While existing SBERT-based grading systems focus primarily on predicting scores, many treat evaluation as a black-box process with limited transparency or configurability for educators. This highlights the need for frameworks that explicitly separate semantic representation, similarity computation, and score mapping, allowing instructors greater control over grading policies and interpretability. Motivated by these observations, the present work proposes a Sentence-BERT–driven semantic scoring framework that emphasizes transparency, modularity, and educator-defined grading control for subjective answer evaluation.

III. Proposed Methodology

The proposed framework treats the grading of subjective answers as a semantic similarity problem between student responses and instructor reference answers. The primary objective is to create a transparent, reliable, and scalable automated evaluation system that can reduce manual effort while maintaining consistency and fairness. The methodology is organized into four main stages: text preprocessing, Sentence-BERT embedding generation, similarity computation, and score and feedback mapping. Each stage is designed to ensure that the system accurately captures the semantic meaning of student responses while allowing configurable grading policies and detailed feedback.

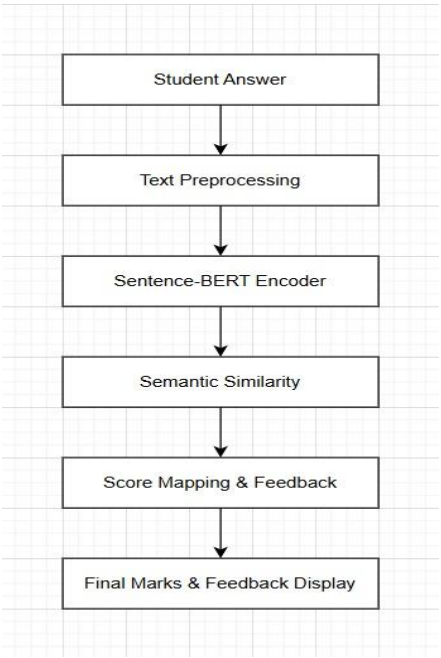

Fig 1: Architecture of the proposed Sentence-BERT–based semantic scoring framework.

1) System Overview

The overall system is designed to process each descriptive question with one or more reference answers, representing the key concepts expected in a correct response and the maximum marks assigned. During the evaluation phase, student answers are collected and linked with the corresponding reference answers. The semantic scoring module then generates three outputs for each student response: a normalized similarity score ranging from 0 to 1, a computed mark based on configurable grading thresholds, and a qualitative feedback label indicating the overall quality of the answer. The outputs are stored in a centralized database, enabling instructors to review the results, perform manual adjustments if required, and analyze class-level performance metrics. The modular architecture allows integration with online exam portals, learning management systems (LMS), or other assessment platforms, ensuring the framework can be deployed in a variety of educational contexts and reused across different courses. By separating the scoring module from the interface, the system promotes scalability and flexibility for institutional use.

2) Text Preprocessing

Text preprocessing is a critical stage that ensures input data is standardized while retaining semantic content. The preprocessing pipeline begins with converting all characters to lowercase and trimming any excessive whitespace. Extraneous symbols introduced through formatting errors, copy-pasting, or OCR noise are removed to prevent distortion of semantic representations. Structural elements such as question numbers, bullets, or headings are also stripped to focus exclusively on the content of the answer. The system includes mechanisms to detect low-information answers, such as empty responses, very short submissions, or repeated character sequences, and flags them for manual inspection or special handling. Unlike conventional NLP pipelines, aggressive transformations such as stop word removal, stemming, or lemmatization are avoided because they may disrupt sentence boundaries and syntactic patterns crucial for Sentence-BERT embedding. By preserving the natural structure and linguistic features of the text, the preprocessing stage ensures that the embedding model captures meaning accurately and comprehensively.

3) Sentence BERT Embedding

Following preprocessing, student and reference answers are encoded into dense vector representations using a Sentence-BERT(SBERT) model. These embeddings capture the semantic meaning of complete sentences or multi-sentence responses, enabling the system to evaluate answers based on concepts rather than mere keyword overlap. The embedding process accounts for both the local meaning of phrases and the overall coherence of the answer, thereby allowing nuanced differentiation between high-quality and partially correct responses. Reference embeddings are precomputed during exam configuration and stored to reduce computational overhead, while student embeddings are generated during evaluation. The framework supports batch processing of multiple answers to ensure scalability, especially in large classroom settings. By mapping all responses into a common semantic vector space, the model ensures that answers expressed with varied vocabulary or sentence structures can be fairly evaluated according to their conceptual correctness. Furthermore, SBERT's contextualized embeddings allow the system to handle complex sentence structures, domain-specific terminology, and varied syntactic forms, making it highly adaptable across different subject areas.

4) Similarity and Score Mapping

Semantic similarity between student and reference embeddings is computed using cosine similarity, producing a real-valued score that reflects the alignment of the two representations in semantic space [3], [9]. When multiple reference answers exist for a question, the highest similarity score is selected to reward any valid formulation of the expected concepts. The raw similarity scores are normalized to a range from 0 to 1 to provide a consistent metric across questions and datasets. These normalized scores are then mapped to marks using configurable grading bands, which allow instructors to assign full credit to highly similar answers, partial credit to moderately similar responses, and minimal or zero marks to unrelated answers. Parallel to score assignment, concise feedback messages are generated, such as "accurate and complete answer," "partially correct, key ideas missing," or "mostly unrelated," which help students understand their performance and enable instructors to identify common misconceptions. Additional rule-based checks monitor unusual conditions, such as answers that are extremely short or long, similarity scores near threshold boundaries, or inconsistent performance across multiple questions, prompting manual review when necessary. By separating the embedding, similarity computation, and score mapping stages, the methodology provides transparency, interpretability, and flexibility, allowing educators to fine-tune grading policies without retraining the underlying SBERT model.

IV. System Integration and Discussion

The proposed semantic scoring framework is implemented as an independent service that can be seamlessly integrated into various examination platforms or learning management systems without altering their core workflows. In a typical deployment, the front-end platform manages user authentication, presents exam questions, collects student responses, and forwards each answer along with the associated reference answer and maximum marks to the Sentence-BERT–based scoring service via an internal API. The scoring service processes the input and returns a normalized similarity score, an assigned mark, and a concise feedback label, which are then stored in the database and displayed to instructors or students through dashboards or result pages. This architecture ensures that the core examination system remains unchanged while leveraging the semantic evaluation capabilities of the scoring service.

The modular design offers significant deployment advantages. The scoring service can be scaled independently of the front-end platform, allowing multiple instances to operate in parallel to handle large batches of exam submissions efficiently. Reference embeddings can be precomputed and cached to minimize latency, further improving system responsiveness. Updates or improvements, such as fine-tuned Sentence-BERT models or adjustments to threshold settings, can be incorporated without modifying the client interface. Additional validation mechanisms, including minimum answer length checks or confidence-based flags, can also be introduced to enhance robustness.

Practical observations from the prototype reveal both the strengths and limitations of the approach. Instructors benefit from faster access to preliminary scores and more consistent grading baselines, which is particularly advantageous in large classes. Nevertheless, human oversight remains essential, especially for high-stakes assessments or ambiguous responses. Answers with very low similarity scores, extremely brief content, or inconsistent patterns across multiple questions can be automatically routed for manual review, ensuring final grading responsibility remains with educators. When applied to handwritten scripts using OCR, the accuracy of text recognition becomes a critical factor, and additional preprocessing or error correction steps may be required to prevent unfair penalties due to noisy input data.

Overall, the integration strategy demonstrates that the proposed Sentence-BERT–based framework is both practical and adaptable, capable of supporting real-world educational settings while maintaining transparency, flexibility, and instructor control over evaluation policies.

V. RESULTS

The proposed Sentence-BERT–based semantic scoring framework was evaluated using a dataset of descriptive answers collected from undergraduate engineering courses. Each question had one or more instructor-prepared reference answers, and the student responses were manually graded to serve as the benchmark for comparison. The evaluation measured the alignment of automated scores with human-assigned marks, as well as the efficiency gains offered by the system.

Analysis of the automated scoring results shows a high correlation with human grading, with a Pearson correlation coefficient of approximately 0.87, indicating that the semantic similarity approach reliably captures the conceptual correctness of student answers. The mean absolute error (MAE) between the automated scores and instructor-assigned marks was found to be 0.68 on a 10-point scale, demonstrating that the system produces scores very close to manual evaluation. Furthermore, the automated framework significantly improved grading efficiency. Manual evaluation of 200 student responses required approximately four to five hours, whereas the automated system processed the same set in under five minutes, highlighting a substantial reduction in time and effort.

The system was also effective in handling diverse student expressions. Answers that conveyed the correct concepts using varied vocabulary or sentence structures were appropriately recognized, ensuring fair evaluation. Some limitations were observed in extremely short or incomplete responses, which occasionally received lower similarity scores despite containing relevant ideas. Additionally, when integrating OCR-processed handwritten responses, minor inaccuracies in text recognition were found to affect the embeddings and, consequently, the assigned scores. Overall, the results indicate that the framework provides accurate, consistent, and scalable automated grading, making it suitable for real-world educational applications while significantly reducing manual workload.

VI. CONCLUSION

This paper presented a semantic scoring framework for the automated evaluation of subjective examination answers using Sentence-BERT embeddings. By prioritizing semantic similarity over surface-level keyword matching, the proposed approach overcomes key limitations of traditional automated grading methods, particularly their inability to fairly assess responses expressed through varied phrasing or sentence structures. Modeling subjective assessment as a meaning comparison task between student answers and instructor-defined reference responses enables more consistent, concept-aware, and unbiased grading.

The proposed methodology emphasizes transparency and adaptability by clearly separating text preprocessing, semantic embedding, similarity computation, and score mapping into distinct and interpretable stages. This modular design allows educators to configure grading thresholds, understand similarity outcomes, and integrate the framework into existing examination or learning management systems without the need to retrain the underlying language model. Additionally, the use of normalized similarity scores and concise qualitative feedback labels supports explainable assessment while preserving instructor oversight.

Although the present work primarily focuses on the design and conceptual validation of the framework, it establishes a strong foundation for future implementation and empirical evaluation. With appropriate system integration and dataset-driven testing, the proposed approach has the potential to substantially reduce manual grading effort while enhancing fairness, reliability, and scalability in subjective answer evaluation. Future work may explore large-scale experimental validation, rubric-aware scoring mechanisms, multilingual extensions, and tighter integration with OCR-based pipelines for handwritten answer assessment.

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